

ABHINAV KAUSHIK

Senior Full-Stack AI Engineer

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SUMMARY

Senior Full-Stack AI Engineer with 8+ years taking LLM and agentic systems from zero to production across Healthcare, Telecom and AIOps, with an MSc in Machine Learning and AI from Liverpool John Moores University. Builds agent loops from the ground up – message arrays, tool calling, context management and prompt caching – and ships them end-to-end, from Python and FastAPI APIs to React and Next.js demo UIs and interactive mockups. Designs evaluations that decide whether a concept holds up, measuring quality, latency and cost with LLM evaluators, tracing and observability, and works AI-assisted with Claude Code and Cursor as the default way of building. Prototypes fast, shows work to real users early, and drops what the evidence does not support. Looking for opportunities in Berlin and willing to relocate. Currently building ThinkTree.AI, a non-profit AI-powered learning platform designed to support students with ADHD and context-switching challenges.

PROFESSIONAL EXPERIENCE

Senior Full-Stack AI Engineer | United Health Group | Bengaluru, India

Nov 2025 – Present

- Developed 21+ automated HTML, TypeScript and Next.js mockup generators on a Node.js stack, turning Jira tickets and PRD requirements into interactive React UI prototypes and clickable demo interfaces that stakeholders reviewed before build, carrying them from POC through to production.
- Engineered a production multi-agent analytics chatbot in LangGraph for AARP medical plans, with tool calling over PySpark, multi-turn context management and a live charting UI, cutting manual analysis by 98%.
- Designed automated evaluation and observability pipelines for Question Answering and Summary agents with LLM evaluators, tracing and failure detection, reducing human-in-the-loop review by 43%.
- Shipped a production document-extraction agent with Claude Opus 4.7 and Amazon Textract, integrated with LIMS via REST API, processing 1,800 lab documents and automating 800 Jira tickets.
- Fine-tuned a cross-encoder on positive and negative pairs to re-rank dense retrieval over AARP survey data, improving retrieval accuracy.

Senior Machine Learning Engineer | Viavi Solutions | Chennai, India

Sep 2024 – Nov 2025

- Prototyped 15+ AI applications from ambiguous client requirements, taking each from idea to working software and demo UI, validating with users, and carrying 7 into production or paid client work while dropping the rest.
- Built an internal RAG-based Project Discovery Assistant in LangGraph, combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions and adopted by internal teams.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace cascading network alarm floods back to the root event, improving semantic alarm classification by 73.5%.

Data Scientist | Viavi Solutions | Bengaluru, India

Nov 2021 – Sep 2024

- Deployed a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, enabling earlier detection and proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | Optum | Gurugram, India

Jun 2021 – Nov 2021

- Implemented a reranker for an existing recommendation system serving individual and child plan recommendations, optimizing renewal suggestions and increasing conversion by 2%.
- Developed a Python and Flask parser that automated extraction and transformation of raw data into structured datasets, feeding downstream analytics and reporting.

Data Engineer | Nippon Data Systems | Gurugram, India

Nov 2019 – May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

Associate Consultant | Saksoft | Gurugram, India

Dec 2017 – Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

PROJECTS

Think Tree – ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Agent Harness and Evaluation Playground

- Built an agent loop from scratch – message arrays, tool calling, context management and prompt caching – with external tools wired in over MCP (Model Context Protocol) and a Next.js and React playground UI to run side-by-side experiments scoring competing agent patterns on answer quality, latency and token cost, so a prototype can be kept, iterated or killed on evidence.

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

Financial Fraud Detection – Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 – Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 – Aug 2017

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM, AI Agents, Agent Loops, Tool Calling, MCP (Model Context Protocol), Context Management, Prompt Caching, Agentic AI, RAG, Agentic RAG, LangGraph, LangChain, Claude, OpenAI, Conversational AI, Chatbots, Multi-Turn Dialogue, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP

Frontend / Full-Stack: React, Next.js, TypeScript, JavaScript, Node.js, HTML, CSS, UI Prototyping, Mockups, Demo UIs, REST API, Microservices

AI-Assisted Development: Claude Code, Cursor, LLM Pair Programming, Rapid Prototyping

Evals / Observability: AI Evaluations, LLM Evals, LLM Evaluators, Agent Tracing, Observability, Latency and Cost Benchmarking, Elasticsearch, Kibana

Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Anomaly Detection, Deep Learning

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Spark SQL, Pydantic, Unix

MLOps / Cloud: MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bitbucket, GCP, Azure, Vercel

Databases: PostgreSQL, TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

LANGUAGES

English: Native

German: A1 – Pursuing

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies